Journal of Clinical Immunology
https://doi.org/10.1007/s10875-019-00726-0

LETTER TO EDITOR

TTC7A Variants Previously Described to Cause Enteropathy Are Observed on a Single Haplotype and Appear Non-pathogenic in Pediatric Inflammatory Bowel Disease Patients

James J. Ashton[1,2] · Enrico Mossotto[1,3] · R. Mark Beattie[2] · Sarah Ennis[1]

Received: 25 July 2019 / Accepted: 24 November 2019
© Springer Science+Business Media, LLC, part of Springer Nature 2019

To the Editor,

Identification of monogenic causes of inflammatory bowel disease (IBD) has increased dramatically over the last 5 years, largely due to the improved interpretation and value offered through the use of whole exome and whole genome sequencing (WES + WGS) [1]. There are now over 80 genes associated with Mendelian causes of IBD, with novel associations being described each year [1]. Despite this, there are significant challenges with interpretation of variants that appear to be deleterious in silico, but lack functional validation [1]. An additional challenge lies with identification of compound heterozygosity within patients.

We read with interest the report from Lawless et al. of compound heterozygote variants in the Tetratricopeptide Repeat Domain 7A (*TTC7A*) gene presenting with enteropathy and a common variable immunodeficiency (CVID) phenotype [2]. *TTC7A* deficiency has been described as a cause of monogenic IBD in over 50 patients, in an autosomal recessive inheritance pattern [3]. Multiple variants have been demonstrated to be causal, presenting with a varied phenotype involving inflammatory, gastrointestinal and immunological manifestations [3]. Typically nonsense variants are fatal within a year of birth, whereas missense mutations may lead to milder disease and patients surviving into adulthood [2, 3]. The role of variation within *TTC7A* in non-monogenic forms of IBD is less clear. *TTC7A* is functionally key as a scaffolding and chaperone protein in processes crucial for normal intestinal epithelial cell development [3]. It may prove that milder missense mutations are risk variants for polygenic forms of IBD, although this has not been substantiated.

We have previously reported monogenic forms of inflammatory bowel disease for our cohort [4]. In this study, we describe the two variants observed in trans by Lawless et al. in five of our patients, always observed in cis. Additionally, through application of a contemporary per gene, per individual, deleteriousness score, GenePy, we assess the evidence for variation in *TTC7A* contributing to polygenic IBD [5].

Patients recruited to the genetics of pediatric IBD study ($n = 401$) at Southampton Children's Hospital underwent WES. All patients were consented and the study had ethics approval from Southampton and South West Hampshire Research Ethics Committee (09/H0504/125). Data were processed through an in-house pipeline. Parental DNA underwent Sanger sequencing to confirm segregation in potential compound heterozygous patients [5]. Variants were annotated using in silico deleteriousness metrics including ClinVar (2018 update) and human mutation genetic database (HGMD Pro 2018). Variant call files (VCFs) for *TTC7A* in 401 patients with IBD were assessed for pathogenic mutations.

Five patients were identified harboring the p.K606R and p.S672P variants (Table 1). Sanger sequencing confirmed all patients had these variants on the same chromosome (inherited maternally or paternally). Neither variant was observed in isolation (without the other) in the remaining 396 patients. In all cases, the parent who transmitted these variants did not have IBD. The brother of patient 1, who also had IBD (Crohn's

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s10875-019-00726-0) contains supplementary material, which is available to authorized users.

✉ Sarah Ennis
s.ennis@southampton.ac.uk

[1] Department of Human Genetics and Genomic Medicine, University of Southampton, Duthie Building (Mailpoint 808), Southampton General Hospital, Southampton SO16 6YD, UK

[2] Department of Paediatric Gastroenterology, Southampton Children's Hospital, Southampton, UK

[3] NIHR Southampton Biomedical Research Centre, University Hospital Southampton, Southampton, UK

Published online: 09 December 2019

Table 1 Genotype and phenotype characteristics of five patients identified harboring the p.K606R and p.S672P *TTC7A* variants

Patient	*TTC7A* variants	gNomad frequency	CADD score	Inherited from	Second *TTC7A* variant?	Second *TTC7A* variant CADD score	Second *TTC7A* variant frequency	Sex	Age at diagnosis	Follow-up duration	Disease	Paris classification	Complications including evidence of immunodeficiency	Medications used	Additional information
1	K606R S672P	0.0022 0.0022	25.6 27.5	Mother	Synonymous G696A:p.-E232E	9.522	0.1778	Male	12.9 years	4.4 years	Ulcerative colitis	E4	None evident	5-ASA Thiopurine	No FH
2	K606R S672P	0.0022 0.0022	25.6 27.5	Father	None	-	-	Female	12.6 years	8.9 years	Crohn's disease	L3 + L4a	None evident	5-ASA Thiopurine	Brother with IBD also heterozygote for *TTC7A* variants
3	K606R S672P	0.0022 0.0022	25.6 27.5	Mother	Synonymous G696A:p.-E232E	9.522	0.1778	Male	15.2 years	2.7 years	Crohn's disease	L1	None evident	Anti-TNF	No FH
4	K606R S672P	0.0022 0.0022	25.6 27.5	Father	Synonymous G696A:p.-E232E	9.522	0.1778	Male	13.4 years	2.8 years	Crohn's disease	L2	None evident	5-ASA Thiopurine	No FH
5	K606R S672P	0.0022 0.0022	25.6 27.5	Mother	Synonymous C198T:p.-S66S	17.63	0.08	Male	15.7 years	1.4 years	Crohn's disease	L3 + L4a	Stricturing disease	Anti-TNF	No FH; Has *NOD2* variant

disease, presenting in childhood), was found to have both the p.K606R and p.S672P variants. The phenotype of these *TTC7A* patients was varied, 4/5 probands had Crohn's disease. One had required surgery during follow-up (mean follow-up time 4.0 years) due to stricturing disease but none of the remaining patients had developed complications and disease was controlled with 5-ASA, thiopurine, or anti-TNF therapy.

The p.K606R and p.S672P variants are rare, both having a gNomad (all genomes) frequency of 0.0022 and an EXAC (non-Finnish European) frequency of 0.0035. Identical frequency suggests inheritance on a common haplotype. Both are predicted to be highly deleterious, possessing CADD scores of 25.6 and 27.5, respectively. The percentage of our patients, as a proportion of IBD patients sequenced, possessing these alleles is 1.2% (5 patients), contrasting to the 1 patient (0.22–0.35%) expected from curated frequency databases. To determine whether an additional variant could be impacting on the function of the wild type allele, patient VCFs were reviewed. Three patients (patients 1, 3, and 4) harbored the common synonymous p.E232E variant (CADD 9.5) and one (patient 5) harbored the synonymous p.S66S variant (CADD 17.63). Both of these are common, minor allele frequency 0.178 and 0.08, respectively, and have modest deleteriousness scores. Segregation analysis was not performed for these variants. Sanger sequencing to confirm segregation unambiguously confirmed heterozygote status and excluded a deletion in the five patients at the site of the p.K606R and p.S672P variants. To further assess the evidence for large deletions and copy number variants (CNV), WES data from patients harboring TTC7A variants were examined in the integrative genome viewer (IGV) and compared with patients not harboring any TTC7A variants, sequencing using the same capture kits. Read depth was calculated across the entire gene and compared with patients without the variants (supplementary Fig. 1). No deletions or CNVs were identified through either method. However, we acknowledge the limitations of WES in the detection of deletions.

Based on the key role *TTC7A* has in regulation of normal intestinal epithelial cell development, we hypothesized that high mutation burden in this gene may contribute to IBD through mutations not classified as casual of a monogenic disorder. We utilized a novel per gene, per individual deleteriousness score, GenePy, to assess whether variation in *TTC7A* was contributing to the risk of developing inflammatory bowel disease but not in a monogenic inheritance pattern [5]. GenePy provides a single score for a gene, for each individual, based on all variants harbored within that gene. Even in genes known to play a role in the development of polygenic disease high scores are uncommon in individual risk genes [5]. Therefore, comparison of extreme scores (top 10%, Mann–Whitney U test) in *TTC7A* between all cases and a set of non-IBD controls allows determination of whether

variation in *TTC7A* was more deleterious in our IBD cohort. Following application of GenePy, mutation burden was identical in cases and controls, with the median of the top 10% of scores being 0.72 in both, $p = > 0.05$. This provides evidence that *TTC7A* variation observed in our cohort was not contributing to polygenic disease development.

The p.K606R and p.S672P variants appear to be most commonly inherited on the same haplotype. The number of patients harboring these heterozygote variants exceeds that expected from both gNomad and EXAC allele frequency databases. However, there is no evidence that a single affected allele would translate into impact on protein expression or function. Based on all previous reports of disease associated with *TTC7A* variants it appears to be a Mendelian disorder with autosomal recessive inheritance, although previous data has indicated a role for heterozygote variants in other genes associated with autosomal recessive monogenic IBD for other genes [3, 4, 6]. Where these variants have been inherited in trans, previous reports have detailed a partial loss-of-function associated with mild CVID and enteropathy phenotype [2]. Interestingly, neither of the parents of the patient described by Lawless et al. possessed the p.K606R and p.S672P haplotype, with the mother having the wild type allele at amino acid position 606 and the father being wild type at position 672.

In this study, we identify five pediatric IBD patients harboring variants on the same chromosome, previously described as disease-causing when inherited in compound heterozygosity. These variants, p.K606R and p.S672P, appear to exist on a relatively rare ancestral haplotype and are seen in 1.2% of our cohort. *In silico* analysis indicates that exonic variation in *TTC7A* was not contributing to disease in this cohort. We find no evidence of increased *TTC7A* mutation burden in the coding regions for IBD patients compared with controls. These data imply that *TTC7A* is only associated with an autosomal recessive Mendelian disorder, with disease occurring in patients who harbor rare, deleterious variants. We were unable to assess the impact of variation in intronic/promoter region, and alteration of *TTC7A* expression or alternative splicing could still contribute to disease pathogenesis. Functional analysis is required to assess the role of rare heterozygote variants, not fulfilling the criteria for causing monogenic disease, in the regulation of intestinal development and inflammation.

Funding Information JJA is funded by an Action Medical Research, research training fellowship and by an ESPEN fellowship. There is no specific funding for this manuscript. This study is supported by the National Institute for Health Research (NIHR) Southampton Biomedical Centre.

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

Ethics Statement All procedures performed in studies involving human participants were in accordance with the ethical standards of Southampton and South West Hampshire Research Ethics Committee (09/H0504/125) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Disclaimer The views expressed are those of the authors and not necessarily those of the NIHR or the Department of Health and Social Care.

References

1. Uhlig HH, Muise AM. Clinical genomics in inflammatory bowel disease. Trends Genet . 2017 [cited 2019 Jan 21];33(9):629–41. Available from: http://www.ncbi.nlm.nih.gov/pubmed/28755896.
2. Lawless D, Mistry A, Wood PM, Stahlschmidt J, Arumugakani G, Hull M, et al. Bialellic mutations in tetratricopeptide repeat domain 7A (TTC7A) cause common variable immunodeficiency-like phenotype with enteropathy. J Clin Immunol. 2017 [cited 2018 Dec 19];37(7):617–22. Available from: http://www.ncbi.nlm.nih.gov/pubmed/28808844.
3. Jardine S, Dhingani N, Muise AM. TTC7A: steward of intestinal health. Cell Mol Gastroenterol Hepatol. 2018 [cited 2018 Dec 19]; Available from: https://doi.org/10.1016/j.jcmgh.2018.12.001.
4. Ashton JJ, Andreoletti G, Coelho T, Haggarty R, Batra A, Afzal NA, et al. Identification of variants in genes associated with single-gene inflammatory bowel disease by whole-exome sequencing. Inflamm Bowel Dis. 2016 [cited 2018 Mar 8];22(10):2317–27. Available from: http://www.ncbi.nlm.nih.gov/pubmed/27537055.
5. Mossotto E, Ashton JJ, O'Gorman L, Pengelly RJ, Beattie RM, MacArthur BD, et al. GenePy - a score for estimating gene pathogenicity in individuals using next-generation sequencing data. BMC Bioinformatics. 2019 16 [cited 2019 May 24];20(1):254. Available from: https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-2877-3
6. Saunders JR, Lehman A, Turvey SE, Pan J, Rajcan-Separovic E, Muise AM, et al. Novel exonic deletions in TTC7A in a newborn with multiple intestinal atresia and combined immunodeficiency. J Clin Immunol. 2019 [cited 2019 Sep 17];39(6):616–9. Available from: http://www.ncbi.nlm.nih.gov/pubmed/31342292.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.